

Affine Schedule Response in a Synthetic Rotated-Surface-Code Fault Model

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Abstract

This note documents a prospective, fail-closed numerical mechanism experiment for an affine schedule response in a synthetic rotated-surface-code fault model. The experiment does not claim a threshold improvement, an asymptotic logical-error exponent change, a compiler-derived pulse model, a hardware advantage, or a reduction of real physical surface-code resources.

Claim Boundary

The ideal logical identity layer is inserted constructively by Stim. The schedule does not generate distinct physical unitary evolutions. Instead, it changes a frozen synthetic schedule-to-fault map: single-data and adjacent-pair fault probabilities are attached to the inserted ideal layer, and PyMatching is constructed from each arm detector error model.

Prospective Result

In the frozen v0.6.0 prospective cohort, specified and frozen before prospective evaluation, constrained schedule optimization reduced the minimum tested distance satisfying the frozen Wilson criterion from 11 to 9 at target logical failure probability $p_L = 0.025$ on all three new prospective seeds. With syndrome rounds equal to distance, this corresponds to a 45.34% active-coordinate qubit-round proxy saving, not a demonstrated reduction in calibrated hardware resources.

Reproducibility

The code repository for this artifact is <https://github.com/papasop/quantum>. The frozen protocol artifact is stored as `results/v0.6.0/protocol.json` and hashes to `fec91e30001712f3d9ac84c0e45a6b70f2d5ae7189d3c9ac6d1096d47505cbf6` under the repository canonical JSON convention. The archived report certificate hash is `2db9620419ac5a7ff64510c65e0d391c4603b6c361fdd8aadd2d9f96165cbc79`. The compressed archival artifact is `results/v0.6.0/report.json.gz`.

Scientific Positioning

This is not an independent prediction validation: the optimization objective is built from the same synthetic schedule-to-fault probability model later injected into Stim. Decoded Monte Carlo outcomes are held out, but the analytic fault map optimized by the flow is also the map injected into Stim; this is not an out-of-model validation.